Gangidi Rasagna Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.61
Sri Chaitanya College, Madhapur, Hyderabad.	Class XII (Telangana Board of Intermediate Education, Hyderabad)	2024	98.50%
Sri Chaitanya High School, Suncity, Hyderabad.	Class X (Board of Secondary Education, Telangana State)	2022	10.00%

POSITIONS OF RESPONSIBILITY

UG Mentor (2025-26) - Sunshine (The counselling cell)

- Mentored the freshers assigned and volunteering in some of the events in the club.

Milan (Creatives) Assistant Co-ordinator , Co-ordinator(2024-26) - Annual inter-hostel fest featuring sports,cultural & tech competitions

- Designed brochures, theme reveals,posters and visual assets using Figma.

SKILLS

Languages: C, Python, Verilog, SQL

Circuit Tools: LTSpice, Altium, TinkerCad

Developer Tools: Git, LaTeX

RELEVANT COURSES

- **Core Theory -** Digital Systems, Analog Electronics, Semiconductor Device Fundamentals, Control Systems, Introduction to programming(C), Signal Processing, Circuits and Network Analysis, Data Base Management system 1, Data Structures and Applications.
- **Core Lab -** Digital Systems Lab, Electronic devices and circuits Lab, Electric Circuits Lab.
- **Mathematics Courses -** Differential Equations and Transform Techniques, Introduction to statistics, Series of Functions, Probability and Random Variables, Vector Calculus.

PROJECTS

6-Bit Half-Flash ADC:

- Designed, Verified and Analysed a 6-bit Half-Flash ADC in LTspice, integrating priority encoder, Flash ADC, and DAC for complete analog-to-digital-to-analog conversion.
- Analyzed propagation delay, DAC settling time, and frequency response to evaluate the converter's dynamic performance.

Mod 100 Up/Down Counter using JK Flip-Flops:

- Designed a 2-digit up/down (MOD-100) counter to track occupancy(0-99), using push buttons for increment/decrement.
- Implemented and verified the same on tinkercad.

Sequence Detector:

- Designed and implemented an overlapping 11011 sequence detector using a Moore Finite State Machine with JK flip-flops, deriving excitation logic through Karnaugh map minimization.
- Developed and verified the complete digital circuit in Tinkercad using combinational logic gates and synchronous sequential design, ensuring correct detection of overlapping input sequences.

Mod-7 Asynchronous Counter Design:

- Designed and implemented a Mod-7 asynchronous ripple counter using T flip-flops derived from JK flip-flops, incorporating NAND gate-based reset logic for modulo-7 operation.
- Verified counter functionality through Tinkercad simulation and CRO waveform analysis, evaluating frequency division and asynchronous reset behavior across successive flip-flop stages.

EXTRACURRICULAR

- Active Member of NSS (National Service Scheme).
- Participated in Plantation drives and Swachh Bharat Cleanliness campaigns.